

Global Asymptotic Stabilization of Large-Scale Hydraulic Networks Using Positive Proportional Controls

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Abstract—An industrial case study involving a large-scale hydraulic network underlying a district heating system subject to structural changes is considered. The problem of controlling the pressure drop across the so-called end-user valves in the network to a designated vector of reference values under directional actuator constraints is addressed. The proposed solution consists of a set of decentralized positively constrained proportional control actions. The results show that the closed-loop system always has a globally asymptotically stable equilibrium point independently on the number of end-users. Furthermore, by a proper design of controller gains the closed-loop equilibrium point can be designed to belong to an arbitrarily small neighborhood of the desired equilibrium point. Since there exists a globally asymptotically stable equilibrium point independently on the number of end-users in the system, it is concluded that structural changes can be implemented without risk of introducing instability. In addition, structural changes can be easily implemented due to the decentralized control architecture.

Index Terms—Decentralized control, hydraulic networks, nonlinear systems, robust control.

I. INTRODUCTION

THIS brief involves a large-scale nonlinear hydraulic network underlying a district heating system. The case study has been put forward by one of the industrial partners involved in the Plug and Play Process Control research project [1]. The case study considers a novel paradigm for the design of district heating systems, in which the diameter of the pipes used in the system is reduced to reduce heat dispersion and thereby the overall energy consumption of the network. On the other hand, the reduced diameter of the pipes increase the pressure gradients across the pipes in the network, and thus the pressure losses throughout the system. This can be compensated for by distributing a number of (pressure boosting) pumps at designated points in the network [2], [3]. Apart from the benefit of reduced heat dispersion this architecture with its multiple pump configuration has the additional benefit of being robust toward pump failures, and furthermore it makes the network structure of the system flexible, which means that subsystems, such as end-user connections, can be arbitrarily added to or removed from the system. The objective of the control system is to regulate the pressure across the so-called end-user valves

in the network to a desired reference level using actuators constrained to positive actuation. The control architecture used in this brief consists of a set of decentralized and constrained proportional control actions for the purpose of practical output regulation. The control actions are decentralized in the sense that the control signal for each end-user connection uses measurements obtained only at said end-user. This has the benefit that the control system is scalable and structural network changes are easy to handle. The result presented here considers the global stability properties of the closed-loop system with the proposed control architecture. Previous results involving the same system and control architecture can be found in [4]–[7] where, in the latter, the control was extended to include integral action. In [4] and [5], it was shown that the use of high gains permits the controllers to achieve semiglobal practical output regulation. That is, for each compact set of initial conditions of the closed-loop system there exist positive proportional gains such that the trajectories of the system is attracted by an arbitrarily small neighborhood of the desired reference point, i.e., stability with respect to an open set. In contrast to the set stability result in [4] and [5], the result in [6], which assumed no actuator constraints, showed that for any positive value of proportional gains, there exists a globally asymptotically stable equilibrium point of the closed-loop system, which, with sufficiently large control gains, is within an arbitrarily small neighborhood of the desired reference point. The results in [7] showed that global asymptotic (and local exponential) output regulation can be proven when no constraints are assumed on the direction of actuation. On the other hand the results in [4] and [5] takes explicitly into consideration the fact that the actuators, which typically consists of centrifugal pumps, in the system are only able to deliver actuation in one direction. The original contribution presented here represents an extension of the results in [6], where the actuator constraint is taken into consideration. The result shows that with the constrained control and arbitrary positive control gain, the closed-loop system still achieves a globally asymptotically stable equilibrium point, in contrast to the set stability result given in [4] and [5]. Furthermore, with sufficiently large control gains, this equilibrium point is within an arbitrarily small neighborhood of the desired reference point. In addition, conditions are given under which the constrained and unconstrained system achieves the same equilibrium point. Since the result is independent on the number of end-users in the system and the equilibrium point is globally asymptotically stable, it can be concluded that structural changes can be introduced to the network while still maintaining stability properties of the closed-loop system. Furthermore, the decentralized nature of the control system,

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makes structural changes easy to implement. The remainder of this brief will proceed as follows. In Section II, the model of the hydraulic network is presented along with the proposed controllers. The closed-loop properties of the system are derived in Section III. Section IV presents the results of experiments and numerical simulations performed on a small-scale laboratory system. Finally, the conclusion is drawn in Section V.

A. Nomenclature

Let $\mathbb{R}^n$ denote the n -dimensional Euclidean space, with the scalar product $\langle \mathbf{a}, \mathbf{b} \rangle$ between two vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$. For a vector $\mathbf{x} \in \mathbb{R}^n$, x_i denotes the i th element of $\mathbf{x}$. Let $M(n, m; S)$ denote the set of $n \times m$ matrices with entries from the set S , and $M(n; S) = M(n, n; S)$. For a matrix $\mathbf{A}$, the notation A_{ij} will be used to denote the entry in the i th row and j th column of $\mathbf{A}$. For a square matrix $\mathbf{A}$, $\mathbf{A} > 0$ means that $\mathbf{A}$ is positive definite. For a square matrix $\mathbf{A}$, $\mathbf{A} = \text{diag}(x_i)$ means that $\mathbf{A}$ has x_i as entries on the main diagonal and zero elsewhere. Throughout the following, C^1 denotes a continuously differentiable function (map), and all functions (maps) introduced will be assumed C^1 . For $X \subseteq \mathbb{R}^n$ and $Y \subseteq \mathbb{R}^n$ a map $\mathbf{f} : X \rightarrow Y$ is called (strict) monotonically increasing if for all $\mathbf{a}, \mathbf{b} \in X$ such that $\mathbf{a} \neq \mathbf{b}$, $\langle \mathbf{a} - \mathbf{b}, \mathbf{f}(\mathbf{a}) - \mathbf{f}(\mathbf{b}) \rangle \geq (>)0$. The open ball with radius r centred in $\mathbf{x} \in \mathbb{R}^n$ is denoted $B_r(\mathbf{x})$, i.e., $B_r(\mathbf{x}) = \{\mathbf{y} \in \mathbb{R}^n \mid |\mathbf{y} - \mathbf{x}| < r\}$. The sphere with radius r and centred in $\mathbf{x} \in \mathbb{R}^n$ is denoted $S_r(\mathbf{x})$, i.e., $S_r(\mathbf{x}) = \{\mathbf{y} \in \mathbb{R}^n \mid |\mathbf{y} - \mathbf{x}| = r\}$.

II. SYSTEM MODEL

The model of the system will here be presented in as much detail as deemed needed for the derivation of the main result. For additional details of the model of the system, the interested reader is referred to [5], where the same system is considered and the model is derived. Models similar to the one presented here can be found in [8]–[12] where fluid networks with capacitive elements (tanks) are considered, and in [13] where cardiovascular systems with flow sources are considered. In contrast, the system considered here contains no capacitive elements and is actuated by pressure sources.

The network consists of m pipe, valve, or pump components, which comprise a single heat source and n end-users and their interconnections. The state and output model of the system is given by

$$\mathbf{J}\dot{\mathbf{q}} = \mathbf{f}(\mathbf{B}^T \mathbf{q}) + \mathbf{u} \quad (1)$$

$$y_i = \mu_i(q_i), \quad i = 1, 2, \dots, n \quad (2)$$

where $\mathbf{q} \in \mathbb{R}^n$ is the vector of independent flow variables, which coincides with the flows through the so-called end-user valves [5]; $\mathbf{B} \in M(n, m; \{-1, 0, 1\})$ is the fundamental cycle matrix with respect to the spanning tree $\mathcal{T}$ of the network graph that has the end-user valves as chords; $\mathbf{J} \in M(n; \mathbb{R})$ with $\mathbf{J} = \mathbf{J}^T$ and $\mathbf{J} > 0$ is due to mass inertia of the fluid in the pipe components; $\mathbf{f}(\cdot)$ is zero in zero and is due to friction in pipe and valve components; the input $\mathbf{u} \in \mathbb{R}^n$ is a linear combination of pump pressure inputs; $\mu_i(\cdot)$ is the pressure

drop across the i th end-user valve and is a strict monotonically increasing function which is zero in zero; and $\mathbf{x} = \mathbf{B}^T \mathbf{q} \in \mathbb{R}^m$ is the vector of flows through all system components.

The map $\mathbf{f}(\cdot)$ is given by $\mathbf{f}(\mathbf{x}) = -\mathbf{B}(\lambda(\mathbf{x}) + \mu(\mathbf{x}))$, where $\lambda_i(x_i)$ is nonzero for pipe components only, for which it is zero in zero and strict monotonically increasing; $\mu_i(x_i)$ is nonzero for valve components only, for which it is zero in zero and strict monotonically increasing.

The input $\mathbf{u}$ to the system is given by $\mathbf{u} = \mathbf{B}\Delta\mathbf{h}_p$, where $\Delta\mathbf{h}_p \in \mathbb{R}^m$ is a vector of pressure inputs with $\Delta h_{p,i}$ nonzero only for pump components. Let $\Delta\mathbf{h}_e$ and $\Delta\mathbf{h}_b$ be the vectors of pressures delivered by the end-user and boosting pumps, respectively, then $\mathbf{u}$ can be written as

$$\mathbf{u} = [\mathbf{I}_n \quad \mathbf{F}] \begin{pmatrix} \Delta\mathbf{h}_e \\ \Delta\mathbf{h}_b \end{pmatrix} \quad (3)$$

where $\mathbf{F}$ is the submatrix of $\mathbf{B}$, which maps boosting pumps to the fundamental cycles. The identity matrix in (3) is due to the fact that each fundamental cycle is associated with an end-user with its own end-user pump [5].

A. Output Regulation Problem

It is desired to control the vector $\mathbf{y}$ of outputs to a vector $\mathbf{r}$ of constant reference values, with $\mathbf{r}$ taking value in a compact set $\mathcal{R} = \{\mathbf{r} \in \mathbb{R}^n \mid 0 < r_m \leq r_i \leq r_M\}$. The strategy adopted here is to use a set of decentralized proportional control actions for the purpose of practical output regulation. The control architecture is decentralized in the sense that the individual control actions are based on locally available information only, namely the pressure measurements obtained at the individual end-users. Furthermore, the control actions take the nonnegativity constraint of the actuators in the system explicitly into consideration, and can be seen as an extension of the control actions introduced in [6]. The constraints reflect the fact that the actuators, which typically are centrifugal pumps, deliver nonnegative pressure only.

To this end, first define the following n variables:

$$\tilde{u}_i = -N_i(y_i - r_i), \quad i = 1, 2, \dots, n \quad (4)$$

where $N_i > 0$ is a proportional control gain.

Then, the input u to the system is given by

$$u_i = \max\{0, \tilde{u}_i\}, \quad i = 1, 2, \dots, n \quad (5)$$

which is obviously nonnegative.

In the following section, the stability properties of the closed-loop system will be analyzed.

III. CLOSED-LOOP STABILITY ANALYSIS

The closed-loop system (1), (2), (4), and (5) is given by

$$\mathbf{J}\dot{\mathbf{q}} = \mathbf{f}(\mathbf{B}^T \mathbf{q}) + \max\{0, -\mathbf{N}(\mu(\mathbf{q}) - \mathbf{r})\} \quad (6)$$

where $\mathbf{N} = \text{diag}(N_i)$; $\mu(\mathbf{q}) = (\mu_1(q_1), \dots, \mu_n(q_n))$; and $\max\{0, \mathbf{z}\} \equiv (\max\{0, z_1\}, \dots, \max\{0, z_n\})$. Since the right hand side of (6) is Lipschitz continuous on any bounded subset of the state space, there exists a unique maximal solution. Rewriting (6) gives

$$\mathbf{J}\dot{\mathbf{q}} = \mathbf{f}(\mathbf{B}^T \mathbf{q}) - \mathbf{N}(\min\{\mathbf{r}, \mu(\mathbf{q})\} - \mathbf{r}) \quad (7)$$

where $\min\{\mathbf{z}, \mathbf{z}'\} \equiv (\min\{z_1, z'_1\}, \dots, \min\{z_n, z'_n\})$.

The steady-state condition of (7) is given by

$$\begin{aligned} \mathbf{N}\mathbf{r} &= -\mathbf{f}(\mathbf{B}^T \mathbf{q}) + \mathbf{N}\min\{\mathbf{r}, \mu(\mathbf{q})\} \Leftrightarrow \\ \mathbf{r} &= -\mathbf{N}^{-1}\mathbf{f}(\mathbf{B}^T \mathbf{q}) + \min\{\mathbf{r}, \mu(\mathbf{q})\}. \end{aligned} \quad (8)$$

Now, define the map $\mathbf{G}_{\mathbf{r}} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ parameterized by the constant vector $\mathbf{r}$ (the vector of reference values) as

$$\mathbf{G}_{\mathbf{r}}(\mathbf{z}) = \min\{\mathbf{r}, \mu(\mathbf{z})\} - \mathbf{N}^{-1}\mathbf{f}(\mathbf{B}^T \mathbf{z}). \quad (9)$$

In what follows, a more specific class of functions $\mu_i(\cdot)$ and $\lambda_i(\cdot)$, which describe the behavior of valves and pipes, respectively, will be used. This class is motivated by the presence of turbulent flows¹ in the system and is given by

$$\mu_i(x_i) = v_i |x_i| x_i, \quad \lambda_i(x_i) = p_i |x_i| x_i \quad (10)$$

where $v_i > 0$ and $p_i > 0$ are parameters of the valves/pipes.

Then, the following proposition shows that for every constant vector $\mathbf{r} \in \mathcal{R}$ of reference values, there exists a unique vector $\mathbf{q}^*$ of independent flows such that

$$\mathbf{q}^* = \mathbf{G}_{\mathbf{r}}^{-1}(\mathbf{r}) \quad (11)$$

which in turn means that for every value of the reference $\mathbf{r}$ belonging to the set $\mathcal{R}$, there exists a unique equilibrium point of the closed-loop system.

Proposition 3.1: For the class of functions defined in (10), and every constant vector $\mathbf{r} \in \mathcal{R}$, the map $\mathbf{G}_{\mathbf{r}} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ defined in (9) is a homeomorphism.

The proof of Proposition 3.1 is left out of this section to maintain the flow of the stability analysis, but it can be found in Appendix. Nonetheless, we remark that the proof concept is similar to the one used in [6], the profound difference is the proof of properness of $\mathbf{G}_{\mathbf{r}}(\cdot)$.

This in turn means that the closed-loop system (7) can be rewritten as

$$\mathbf{J}\dot{\mathbf{q}} = \mathbf{f}(\mathbf{B}^T \mathbf{q}) - \mathbf{f}(\mathbf{B}^T \mathbf{q}^*) - \mathbf{N}(\mathbf{g}(\mathbf{q}) - \mathbf{g}(\mathbf{q}^*)) \quad (12)$$

where $\mathbf{g}(\mathbf{q}) = (g_1(q_1), \dots, g_n(q_n))$, with the functions $g_i : \mathbb{R} \rightarrow \mathbb{R}$ given by $g_i(x) = \min\{r_i, \mu_i(x)\}$. Since $\mu_i(\cdot)$ is strict monotonically increasing and $\min\{r_i, \cdot\}$ is monotonically increasing, it follows that $g_i(\cdot)$ is a monotonically increasing function, that is

$$\langle x - x', g_i(x) - g_i(x') \rangle \geq 0 \quad \forall x \neq x'. \quad (13)$$

Furthermore, the following has been shown in [6]:

$$\langle \mathbf{z} - \mathbf{z}', \mathbf{f}(\mathbf{B}^T \mathbf{z}) - \mathbf{f}(\mathbf{B}^T \mathbf{z}') \rangle < 0 \quad \forall \mathbf{z} \neq \mathbf{z}'. \quad (14)$$

Now, the main result of this section can be stated.

Proposition 3.2: Let the map $\mathbf{G}_{\mathbf{r}} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be given by (9), the functions $\mu_i : \mathbb{R} \rightarrow \mathbb{R}$, and $\lambda_i : \mathbb{R} \rightarrow \mathbb{R}$ be given by (10) and $\mathbf{q}^*$ be given by (11), then $\mathbf{q}^*$ is the globally asymptotically stable equilibrium point of the closed-loop system (1), (2), (4), and (5).

¹Note that since the main motivation for the new system paradigm is a reduction of the diameter of the pipes used, the likelihood for turbulent flow increases.

Proof: Define the change of coordinates $\tilde{\mathbf{q}} = \mathbf{q} - \mathbf{q}^*$ and the Lyapunov function candidate $V : \mathbb{R}^n \rightarrow \mathbb{R}$ given by $V(\tilde{\mathbf{q}}) = 1/2 \langle \tilde{\mathbf{q}}, \mathbf{J}\tilde{\mathbf{q}} \rangle$. The time derivative of $V(\cdot)$ is

$$\begin{aligned} \frac{d}{dt} V(\tilde{\mathbf{q}}) &= \langle \tilde{\mathbf{q}}, \mathbf{J}\dot{\tilde{\mathbf{q}}} \rangle \\ &= \left\langle \mathbf{q} - \mathbf{q}^*, \mathbf{f}(\mathbf{B}^T \mathbf{q}) - \mathbf{f}(\mathbf{B}^T \mathbf{q}^*) - \mathbf{N}(\mathbf{g}(\mathbf{q}) - \mathbf{g}(\mathbf{q}^*)) \right\rangle. \end{aligned} \quad (15)$$

Since $\mathbf{N}$ is diagonal with entries N_i it follows:

$$\begin{aligned} \langle \mathbf{q} - \mathbf{q}^*, \mathbf{N}(\mathbf{g}(\mathbf{q}) - \mathbf{g}(\mathbf{q}^*)) \rangle \\ = \sum_{i=1}^n N_i (q_i - q_i^*) (g_i(q_i) - g_i(q_i^*)). \end{aligned} \quad (16)$$

From (13), (14), and the fact that $N_i > 0$, the following inequality is true:

$$\frac{d}{dt} V(\tilde{\mathbf{q}}) \leq \left\langle \mathbf{q} - \mathbf{q}^*, \mathbf{f}(\mathbf{B}^T \mathbf{q}) - \mathbf{f}(\mathbf{B}^T \mathbf{q}^*) \right\rangle < 0 \quad \forall \mathbf{q} \neq \mathbf{q}^*. \quad (17)$$

From this it is concluded that $\mathbf{q}^*$ is the globally asymptotically stable equilibrium point of the closed-loop system [14, Th. 4.2]. ■

Remark 3.1: In [6], the situation without constraints on the direction of actuation was considered. Here, it was found that the closed-loop equilibrium is associated with the map $\mathbf{F} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ given by

$$\mathbf{F}(\mathbf{z}) = \mu(\mathbf{z}) - \mathbf{N}^{-1}\mathbf{f}(\mathbf{B}^T \mathbf{z}). \quad (18)$$

Using $\mathbf{F}(\cdot)$, the globally asymptotically stable equilibrium point $\hat{\mathbf{q}}$ of the closed-loop system in the unconstrained case is then given by $\hat{\mathbf{q}} = \mathbf{F}^{-1}(\mathbf{r})$. Taking the difference between the maps $\mathbf{G}_{\mathbf{r}}(\cdot)$ and $\mathbf{F}(\cdot)$ gives

$$\begin{aligned} \mathbf{G}_{\mathbf{r}}(\mathbf{z}) - \mathbf{F}(\mathbf{z}) \\ = \min\{\mathbf{r}, \mu(\mathbf{z})\} - \mathbf{N}^{-1}\mathbf{f}(\mathbf{B}^T \mathbf{z}) - (\mu(\mathbf{z}) - \mathbf{N}^{-1}\mathbf{f}(\mathbf{B}^T \mathbf{z})) \\ = \min\{\mathbf{r}, \mu(\mathbf{z})\} - \mu(\mathbf{z}). \end{aligned} \quad (19)$$

This difference evaluated in the point $\hat{\mathbf{q}}$ is

$$\mathbf{G}_{\mathbf{r}}(\hat{\mathbf{q}}) - \mathbf{F}(\hat{\mathbf{q}}) = \min\{\mathbf{r}, \mu(\hat{\mathbf{q}})\} - \mu(\hat{\mathbf{q}}). \quad (20)$$

The above shows that if $\mu_i(\hat{q}_i) \leq r_i$ for every $i = 1, 2, \dots, n$, then $\mathbf{G}_{\mathbf{r}}(\hat{\mathbf{q}}) - \mathbf{F}(\hat{\mathbf{q}}) = 0$ and consequently that the same equilibrium point is attained in both the constrained and unconstrained case.

Remark 3.2: That $\mathbf{G}_{\mathbf{r}}(\cdot)$ is a homeomorphism is a sufficient but not necessary condition for the existence of a unique equilibrium point. The important point is that the vector $\mathbf{r}$ of reference values is in the image of $\mathbf{G}_{\mathbf{r}}(\cdot)$. Then, the result follows from the fact that $\mathbf{G}_{\mathbf{r}}(\cdot)$ is a homeomorphism onto its image (Appendix). However, showing that a point belongs to the image of a function may be a daunting task. Therefore, by requiring the maps $\mu(\cdot)$ and $\lambda(\cdot)$ to be of the class (10), one can further show that $\mathbf{G}_{\mathbf{r}}(\cdot)$ is a proper map and thus a homeomorphism. As a consequence, $\mathbf{r}$ is guaranteed to be in the image of $\mathbf{G}_{\mathbf{r}}(\cdot)$.

Proposition 3.2 shows that the closed-loop system always has a globally asymptotically stable equilibrium point regardless of how the controller gains $N_i > 0$ are chosen.

On the other hand, the result does not show which steady-state output regulation error can be achieved with the proposed control actions. However, the results in [4] and [5] show that for all initial conditions belonging to some compact set, say $\mathcal{Q}$, there exist sufficiently large controller gains such that the steady-state output regulation error can be made arbitrarily small. Since Proposition 3.2 guarantees the existence of such a compact set $\mathcal{Q}$, the result shows that the chosen control strategy combined with large gains allows global practical output regulation. This can be summarized in the following proposition:

Proposition 3.3: For every $\gamma > 0$, there exist proportional gains $N_i^* > 0$ such that the closed-loop system (6) with the class of functions given by (10) and gains $N_i > N_i^*$ has a global asymptotically stable equilibrium point $\mathbf{q}^*$ which fulfils

$$\mu(\mathbf{q}^*) \in B_\gamma(\mathbf{r}).$$

Proof: The proof of Proposition 3.3 is done by combining Proposition 3.2 for the existence of the global asymptotically stable equilibrium point $\mathbf{q}^*$ and [4, Proposition 2] for the existence of gains N_i^* . ■

Since the results presented here are independent on the number n of end-users in the system, it is concluded that end-users can be added to or removed from the system while still maintaining the stability properties of the closed-loop system, given that such structural changes occur sporadically. However, to maintain the same steady-state output regulation error, it may be necessary to redesign the gains of the controllers.

In the preceding, it has been established that there exists a set of proportional control actions which are nonnegative and such that a globally asymptotically stable equilibrium point of the closed-loop system is attained. However, as seen in (3) the control input $\mathbf{u}$ to the system consists of a linear combination of the actual actuator inputs. Now, the question arise if there exists a mapping from the nonnegative control signals to a set of nonnegative actuator inputs. First, it can be noted that the matrix $[\mathbf{I}_n \ \mathbf{F}]$ in (3) is full rank and therefore has a generalized inverse, which can be used as a mapping. However, this mapping does in general not guarantee the actuator inputs to be nonnegative. A mapping that guarantees nonnegative control signals being mapped to nonnegative actuator inputs is given in [15]. The construction of the mapping is based on graph-theoretical reflections, and the interested reader is referred to [15] for additional information.

IV. EXPERIMENTAL AND NUMERICAL RESULTS

The closed-loop system has been tested using experiments on a special laboratory setup² and numerical simulations. The setup emulates a small scale district heating system. However, due to physical constraints imposed by the size of the setup, the dynamics are 5–10 times faster than what would be expected in a real system. A picture of the setup is shown in Fig. 1.

The laboratory setup consists of four end-users, and the diagram of the corresponding hydraulic network is shown in



Fig. 1. Laboratory setup used in experiments and simulations.

Fig. 2. Referring to the figure, the end-users in the system are comprised of the connections $\{c_9, c_{10}, c_{11}\}$, $\{c_{27}, c_{28}, c_{29}\}$, $\{c_{23}, c_{24}, c_{25}\}$, and $\{c_{19}, c_{20}, c_{21}\}$, with the electronically controlled valves $c_{10}, c_{28}, c_{24}, c_{20}$ modeling the pressure drop across the primary side of the heat exchangers at the end-users, and the valve c_{17} modeling the pressure drop across the secondary side of the heat exchanger at the heat source. In the configuration of the setup, the valves c_4 and c_{14} are closed, and the measured outputs are limited to signals $y_3 = dp_3$ and $y_4 = dp_4$, which are the differential pressure measurements across the valves c_{24} and c_{20} , respectively. The input is limited to Pumps 4, 5, and 6 (components c_{23} , c_{19} , and c_1). This scenario corresponds to the two end-user case.

A. Experimental Results

The control signals u_i for $i = 3, 4$ [see (3)] are given as

$$u_3 = \Delta h_{p,1} + \Delta h_{p,23}, \quad u_4 = \Delta h_{p,1} + \Delta h_{p,19}. \quad (21)$$

The control signal u_i is calculated by the proportional control based on the measurement dp_i and then distributed to the pumps in the system based on the mapping provided in [15] using the following rule:

$$\begin{aligned} \Delta h_{p,1} &= 0.7 \min\{u_3, u_4\} \\ \Delta h_{p,23} &= u_3 - \Delta h_{p,1} \\ \Delta h_{p,19} &= u_4 - \Delta h_{p,1}. \end{aligned} \quad (22)$$

In the experiment, a step response is obtained by changing the reference value from 0.2 (bar) to 0.3 (bar) at time 50 s and back at time 250 s. This scenario has been carried out for different values of the gain matrix $\mathbf{N}$, in particular for $\mathbf{N} = \mathbf{I}_2$, $\mathbf{N} = 2\mathbf{I}_2$, $\mathbf{N} = 3\mathbf{I}_3$, and $\mathbf{N} = 4\mathbf{I}_2$. The results of the experiments can be found in Fig. 3, where it is seen that the system attains a steady state for all values of the reference $\mathbf{r}$ and gain matrix $\mathbf{N}$. In particular, it can be noticed that the same steady-state value of y_i is attained at time 50 and 300 s for all values of $\mathbf{N}$. This is related to the fact that the map $\mathbf{G}_r(\cdot)$ being a homeomorphism and consequently the steady-state $\mathbf{q}^*$ unique. Lastly, it can be noticed that increasing the gain will decrease the output regulation error.

²See [5] for additional details on the setup.

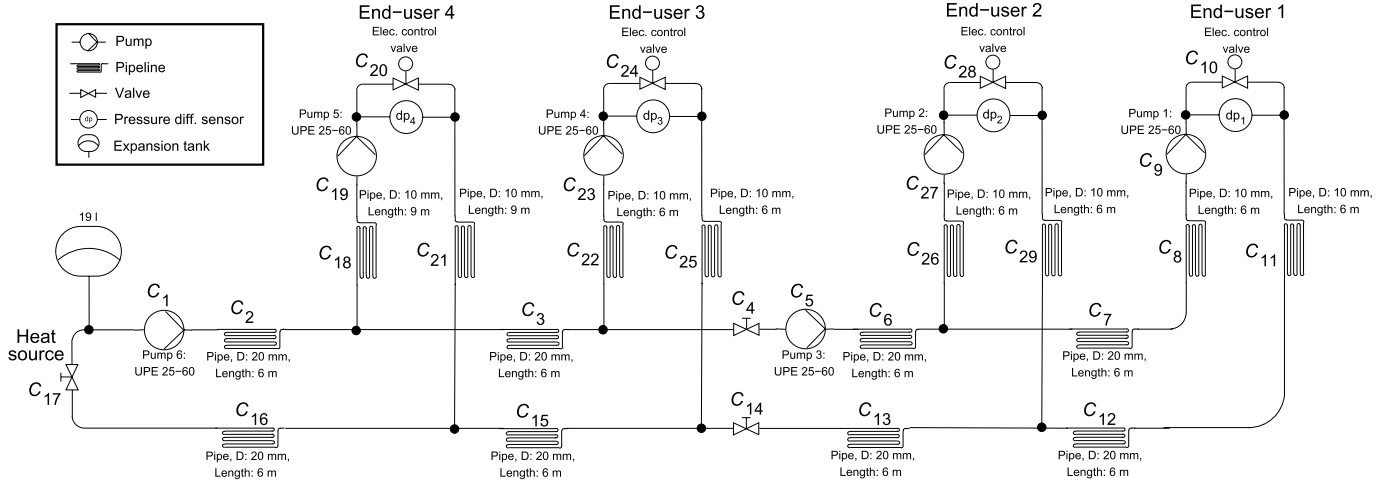


Fig. 2. Hydraulic network diagram for the laboratory setup in Fig. 1.

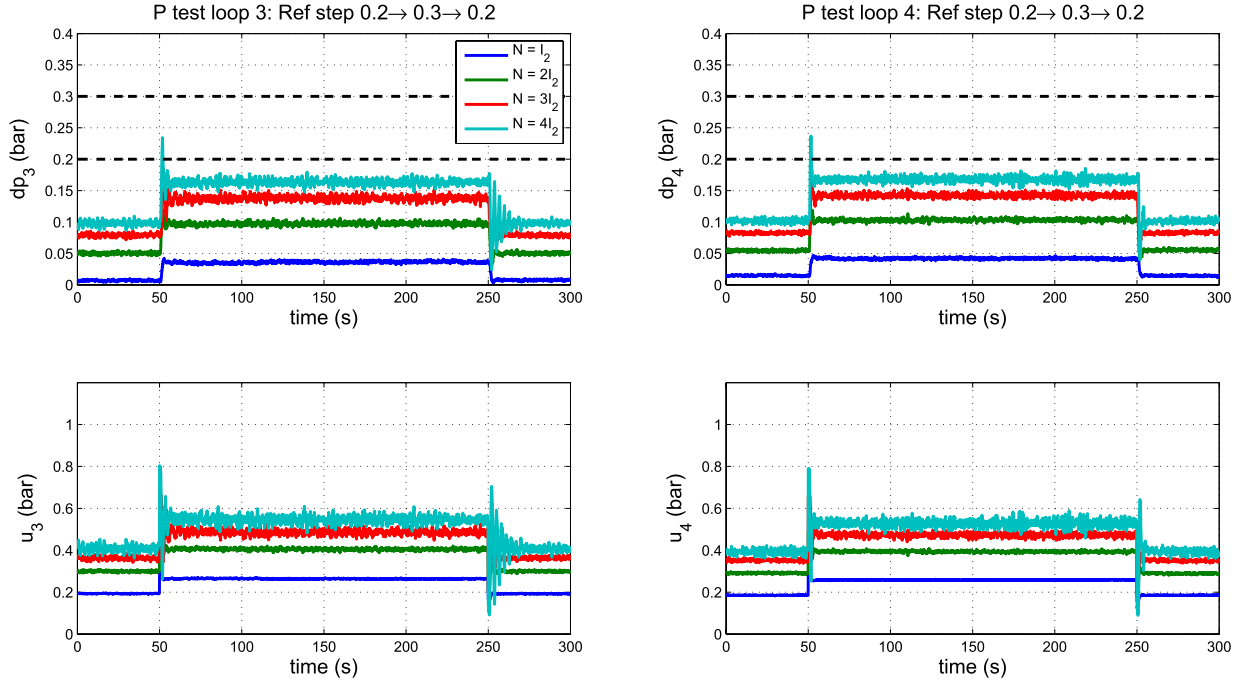


Fig. 3. Results of the experiments conducted on the laboratory setup in Fig. 1. The experiments have been restricted to the end-user connections 3 and 4. The top plots show the pressure measurements dp_3 and dp_4 . The dashed lines indicate the two reference values used to obtain a step response. The bottom plots show the control signals u_3 and u_4 .

B. Numerical Results

The closed-loop system has also been tested using numerical simulations. The system used in the simulations is the four end-user system shown in Fig. 2. The purpose of the simulations is both to verify that the closed-loop system attains a steady state as suggested by Proposition 3.2 and furthermore, to verify that the closed-loop system is able to handle structural changes while maintaining its closed-loop stability. The latter is a feature guaranteed by the decentralized nature of the controller and the fact that Proposition 3.2 is independent on the number of end-users in the system. Finally, the simulation is executed with two different values of gains N_i to illustrate that increasing the gains will decrease the output regulation error.

The class of functions $\mu_i(\cdot)$ and $\lambda_i(\cdot)$ describing the behavior of valves and pipes, respectively, which have been used in the simulations are the ones given in (10). The parameters used in the simulations are: $J_{11} = 0.8488$, $J_{12} = J_{21} = 0.3183$, $J_{22} = J_{44} = 0.7427$, $J_{13} = J_{23} = J_{32} = J_{31} = 0.2122$, $J_{33} = 0.6366$, $J_{14} = J_{24} = J_{34} = J_{43} = J_{42} = J_{41} = 0.1061$, $p_2 = p_3 = p_6 = p_7 = p_{12} = p_{13} = p_{15} = p_{16} = 0.0141$, $p_8 = p_{11} = p_{22} = p_{25} = p_{26} = p_{29} = 0.4503$, $p_{18} = p_{21} = 0.6755$, $v_{10} = v_{28} = v_{24} = v_{20} = 0.005$, $v_{17} = 0.0013$, $\mathbf{r} = 0.2\mathbf{1}_4$, and controller gain matrices $\mathbf{N} = 2\mathbf{I}_4$ and $\mathbf{N} = 20\mathbf{I}_4$.

A scenario has been simulated where the end-user connection consisting of components $\{c_9, c_{10}, c_{11}\}$ is removed from the system at time 50 s and reinserted in the system

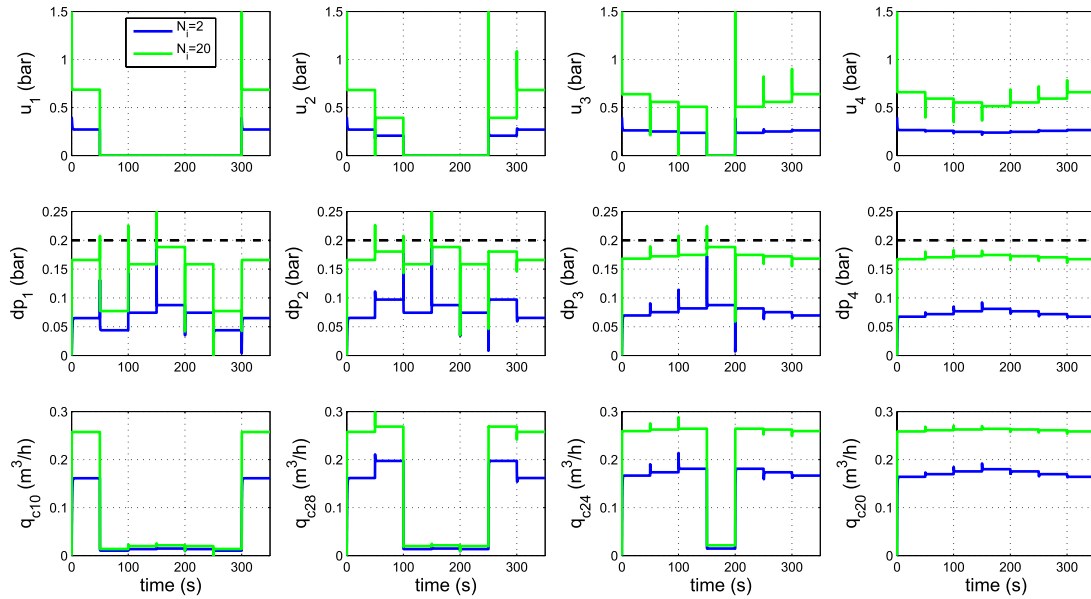


Fig. 4. Result of the simulation of the four end-user system in Fig. 2 in closed loop with the proportional control actions. The figure shows the control inputs u_1 , u_2 , u_3 , and u_4 , the controlled variables dp_1 , dp_2 , dp_3 , and dp_4 , and the flow through valves c_{10} , c_{28} , c_{24} , and c_{20} . The dashed lines in the middle plots indicate the reference output. The end-user connection consisting of components $\{c_9, c_{10}, c_{11}\}$ is removed from the system at time 50 s and later reinserted into the system at time 300 s; the end-user connection consisting of components $\{c_{27}, c_{28}, c_{29}\}$ is removed from the system at time 100 s and reinserted into the system at time 250 s; the end-user connection consisting of components $\{c_{23}, c_{24}, c_{25}\}$ is removed from the system at time 150 s and reinserted into the system at time 200 s.

at time 300 s; the end-user connection consisting of components $\{c_{27}, c_{28}, c_{29}\}$ is removed from the system at time 100 s and reinserted into the system at time 250 s; the end-user connection consisting of components $\{c_{23}, c_{24}, c_{25}\}$ is removed from the system at time 150 s and reinserted into the system at time 200 s. The removal of an end-user connection is done by changing the hydraulic resistance parameter (v_i , $i = 10, 28, 24, 20$) of the end-user valve to a large value ($0.005 + 410$) with the result that the flow through the valve is reduced to a value close to zero. Furthermore, some logic has been introduced to disable the control signal for the end-user whenever this large resistance value is used.

The result of the simulation of the system can be seen in Fig. 4. The figure shows the control inputs from the proportional control actions, the controlled variables and the flows through the end-user valves. As it is evident from the figure, the closed-loop system attains a steady state in all four cases; that is, when, respectively, four, three, two, and one end-users are connected to the system. Furthermore, for the blue plots it can be seen that a steady state of $\mathbf{q}^*(50) = \mathbf{q}^*(350) = (q_{c10}, q_{c28}, q_{c24}, q_{c20}) = (0.1612, 0.1616, 0.1658, 0.1632)$ is attained when four end-users are connected; a steady state of $\mathbf{q}^*(100) = \mathbf{q}^*(300) = (q_{c28}, q_{c24}, q_{c20}) = (0.2044, 0.1726, 0.1686)$ is attained when three end-users are connected; a steady state of $\mathbf{q}^*(150) = \mathbf{q}^*(250) = (q_{c24}, q_{c20}) = (0.1808, 0.1750)$ is attained when two end-users are connected; and a steady state of $q^*(200) = q_{c20} = 0.1803$ is attained when one end-user is connected. Similar behavior is observed for the green plots. That the system attains the same steady state for a particular combination of end-users when all other system parameters remain unchanged is not a coincidence. In fact, this behavior is

expected from the knowledge that the mapping $\mathbf{G}_r(\cdot)$ given by (9) is a homeomorphism. As a consequence, the steady-state $\mathbf{q}^*$ is unique for a particular choice of $\mathbf{r}$, with $\mathbf{q}^*$ given by (11).

A feature also apparent from Fig. 4 is that the steady-state $\mathbf{q}^*$ fulfils $\mu_i(q_i^*) \leq r_i$, and as pointed out in Remark 3.1 this means that in this particular case the closed-loop system attains the same steady state as the system without actuator constraints. Lastly, it can be observed that the steady-state output regulation error is reduced when the gain is increased.

The qualitative difference between Figs. 3 and 4 (no overshoot and oscillations in simulations) is due to the fact that the dynamics of the experimental setup are much faster than in a real-life hydraulic network. Since the dynamics of the actuators installed in the laboratory equipment are the same as in a real system (and hence much slower than the dynamics of the laboratory setup), there is a large delay in the control input, which results in the difference between experimental results and numerical simulations.

V. CONCLUSION

The problem of controlling a set of measured pressures in a large-scale hydraulic network to a vector of desired reference values was considered. The problem is complicated by directional constraints on the actuators used in the system. The class of hydraulic networks examined includes the ones underlying a district heating system with a novel and flexible network structure allowing for structural changes, such as the addition or removal of end-users. A control architecture consisting of a set of decentralized proportional control actions constrained to nonnegative values was considered for accommodating the output regulation problem. The decentralized

nature of the control architecture makes structural changes easy to implement in the control.

The main result of this brief shows that the closed-loop system with the proposed control actions always has a globally asymptotically stable equilibrium point, which differs from the desired equilibrium point. This difference decreases with increasing control gains. Since there exists an equilibrium point of the closed-loop system which is globally asymptotically stable irrespective of the number of end-users it is concluded that end-users can be added to and removed from the system and still maintain the stability properties. However, to keep the same level of performance it may be necessary to redesign the gains of the controller.

The analytic result was supported by simulations performed on a simple system consisting of four end-users and by experiments conducted on a laboratory setup.

APPENDIX

PROOF OF PROPOSITION 3.1

This appendix will provide a proof of Proposition 3.1. Following [16], the aim is to show that: 1) $\mathbf{G}_r(\cdot)$ is a local homeomorphism and 2) $\mathbf{G}_r(\cdot)$ is a proper map. Toward that end, first define the map $\tilde{\mathbf{G}}_r : \mathbb{R}^n \rightarrow \mathbb{R}^n$ as

$$\tilde{\mathbf{G}}_r(\mathbf{z}) = \mathbf{N} \min\{\mathbf{r}, \boldsymbol{\mu}(\mathbf{z})\} - \mathbf{f}(\mathbf{B}^T \mathbf{z}). \quad (23)$$

From the monotonicity properties of $\boldsymbol{\mu}(\cdot)$ and (14), it follows from [17, Lemma 3.1] that $\mathbf{G}_r(\cdot)$ is a homeomorphism onto its image. Now, it can be shown that 1) $\mathbf{G}_r(\cdot)$ is a local homeomorphism: since $\mathbf{G}_r(\mathbf{z}) = \mathbf{N}^{-1} \tilde{\mathbf{G}}_r(\mathbf{z})$ and $\mathbf{N}^{-1}$ is nonsingular, it follows that $\mathbf{G}_r(\cdot)$ is a local homeomorphism. Next, it is shown that 2) $\mathbf{G}_r(\cdot)$ is a proper map.

Lemma A.1: For the class of functions $\mu_i : \mathbb{R} \rightarrow \mathbb{R}$ and $\lambda_i : \mathbb{R} \rightarrow \mathbb{R}$ defined in (10), and every constant vector $\mathbf{r} \in \mathcal{R}$, the map $\mathbf{G}_r : \mathbb{R}^n \rightarrow \mathbb{R}^n$ defined in (9) is proper.

Proof: Define $\mathbf{F} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ as $\mathbf{F}(\mathbf{z}) = \boldsymbol{\mu}(\mathbf{z}) - \mathbf{N}^{-1} \mathbf{f}(\mathbf{B}^T \mathbf{z})$, and rewrite $\mathbf{G}_r(\cdot)$ as

$$\mathbf{G}_r(\mathbf{z}) = \min\{\mathbf{r}, \boldsymbol{\mu}(\mathbf{z})\} - \mathbf{N}^{-1} \mathbf{f}(\mathbf{B}^T \mathbf{z}) + \boldsymbol{\mu}(\mathbf{z}) - \boldsymbol{\mu}(\mathbf{z}) \quad (24)$$

$$= \mathbf{F}(\mathbf{z}) + \min\{\mathbf{r}, \boldsymbol{\mu}(\mathbf{z})\} - \boldsymbol{\mu}(\mathbf{z}) \quad (25)$$

$$= \mathbf{F}(\mathbf{z}) + \min\{\mathbf{r} - \boldsymbol{\mu}(\mathbf{z}), 0\}. \quad (26)$$

For the function class (10), the following homogeneity properties have been shown in [6]:

$$\mathbf{F}(a\mathbf{z}) = a|\mathbf{F}(\mathbf{z})|, \quad \boldsymbol{\mu}(a\mathbf{z}) = a|\boldsymbol{\mu}(\mathbf{z})| \quad (27)$$

for every $a \in \mathbb{R}$.

Now, from (26), it follows that for every $\varepsilon > 0$, there exists $A > 0$ such that

$$|\mathbf{G}_r(a\mathbf{z}) - a^2 \mathbf{G}_r(\mathbf{z})| < \varepsilon \quad \forall a > A. \quad (28)$$

Observe that $\mathbf{G}_r(0) = 0$ (since $r_i > 0$) and that $\mathbf{G}_r(\cdot)$ is a homeomorphism onto its image. Thus, there exist some open set $\mathcal{U} \subset \mathbb{R}^n$ with $0 \in \mathcal{U}$ and some open set $\mathcal{V} \subset \mathbb{R}^n$ with $0 \in \mathcal{V}$ such that $\mathbf{G}_r : \mathcal{U} \rightarrow \mathcal{V}$ is a homeomorphism. Furthermore, let $r_{\mathcal{U}}$ be such that the sphere $S_{r_{\mathcal{U}}}(0) \subset \mathcal{U}$.

Define the sequence $\{\mathbf{z}^j\}_{j \in \mathbb{N}}$ such that $\lim_{j \rightarrow \infty} |\mathbf{z}^j| = \infty$ and $\mathbf{z}^j \neq 0$. For every $\mathbf{z}^j$, there exists some pair $a^j > 0$ and $\mathbf{x}^j \in S_{r_{\mathcal{U}}}(0)$ such that $\mathbf{z}^j = a^j \mathbf{x}^j$. From this it follows that $\lim_{j \rightarrow \infty} a^j = \infty$. From (28), for every $\varepsilon > 0$ there exists $J(\varepsilon)$ such that for every $j > J(\varepsilon)$

$$\begin{aligned} |\mathbf{G}_r(\mathbf{z}^j)| &= |\mathbf{G}_r(a^j \mathbf{x}^j)| \geq (a^j)^2 |\mathbf{G}_r(\mathbf{x}^j)| - \varepsilon \\ &\geq (a^j)^2 \min_{\mathbf{x} \in S_{r_{\mathcal{U}}}(0)} |\mathbf{G}_r(\mathbf{x})| - \varepsilon. \end{aligned} \quad (29)$$

Since

$$\mathbf{G}_r(\mathbf{x}) \neq 0 \quad \forall \mathbf{x} \in S_{r_{\mathcal{U}}}(0) \quad (30)$$

from [16, Lemma 1] the thesis follows. ■

With the preliminaries above it is possible to prove Proposition 3.1.

Proof: Since $\mathbf{G}_r(\cdot)$ is a local homeomorphism, the thesis follows from Lemma A.1 and the theorem in [16]. ■

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